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Palaeomagnetic and geochemical evaluation of a terrane-linking ignimbrite: evidence for the relative position of the Grampian and Midland Valley terranes in late Silurian time

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Abstract – The proposed correlation of Lower Old Red Sandstone dacitic ignimbrites between the Northern Midland Valley and Grampian terranes of central Scotland has been investigated in a combined palaeomagnetic and geochemical study. Statistically indistinguishable late Silurian palaeolatitudes (Grampian terrane 27.0° S ± 3.5°; Northern Midland Valley terrane 22.9° S ± 1.8°), comparable remagnetization histories, a shared 'normal' polarity for the primary magnetization and equivalent gyroremanence acquisition, favour correlation across the Highland Boundary Fault. Similar chemistry for biotite phenocrysts further corroborates correlation which was initially made on the basis of a similar field appearance and petrography.

This correlation, together with evidence from provenance studies, suggests that the separation of the Northern Midland Valley and Grampian terranes inferred for the Ordovician period was largely removed by late Silurian times, with scope for relative displacements of the order of only tens of kilometres after Lower Old Red Sandstone deposition.

1. Introduction

The displaced terrane hypothesis has been suggested to explain many aspects of the geology of the Caledonian–Appalachian system (e.g. Hutton, 1987; Bluck, 1985). Palaeomagnetic tests of this hypothesis can be divided into two categories: those concerning the former separation of terranes (e.g. Van der Voo & Johnson, 1985; Trench, Bluck & Watts, 1988) and those which relate to the timing of their subsequent amalgamation and suturing. This study is of the latter type, and concerns the late displacement history of terranes juxtaposed by the Highland Boundary Fault (HBF) in central Scotland.

The HBF juxtaposes Precambrian Dalradian metamorphic rocks belonging to the Grampian terrane against the Ordovician Highland Border Complex and its unconformable Lower Old Red Sandstone (ORS) cover. We adopt the term Northern Midland Valley (NMV) terrane here to refer to the Highland Border Complex and its hidden extension beneath the Devonian and Carboniferous cover, recognizing that the Midland Valley itself may conceal important terrane boundaries.

The Grampian and NMV terranes have demonstrably unrelated geological histories during Ordovician time and did not become unambiguously linked until late Devonian time (Bluck, 1985; Fig. 1). Prior to this link, there may have been substantial, and presently unconstrained, strike-slip movements between the two terranes, and such displacements could have persisted until as late as mid-Devonian times (e.g. Hutton, 1987).

Shared structural and/or metamorphic histories, stitching plutons, sediment provenance and stratigraphic overstep can be used to identify the proximity of two terranes. In the absence of stitching plutons, and with ambiguous structural evidence (Johnson & Harris, 1967; Harte *et al.* 1984), the timing of displacements on the HBF and its precursors have been constrained using provenance (Haughton, Rogers & Halliday, 1989) and stratigraphic sediment overstep (e.g. George, 1960; Hutton, 1987). Several authors have drawn attention to the possible overstep of Lower ORS rocks from the NMV to the Grampian terrane. The Lower ORS of the northern Midland Valley comprises a thick succession of conglomerates, sandstones and mudstones, together with important intercalations of volcanic rock.

Although Lower ORS age conglomerates, sandstones and volcanic rocks locally rest on the Grampian terrane, Bluck (1984) drew attention to the disparate composition and dispersal patterns of conglomerates within the respective sequences. In addition there is no locality where the Lower ORS can be seen to overstep the actual faulted contact between Dalradian and Highland Border Complex lithologies. Given the likelihood of reactivation of existing structures, many terrane boundary overstep sequences may suffer protracted histories of later displacement. Rigorous methods allowing correlation of faulted successions are therefore required to establish whether oversteps are truly terrane-linking, or whether two superficially similar successions have been juxtaposed by later terrane displacements.

In the present study we address the possible

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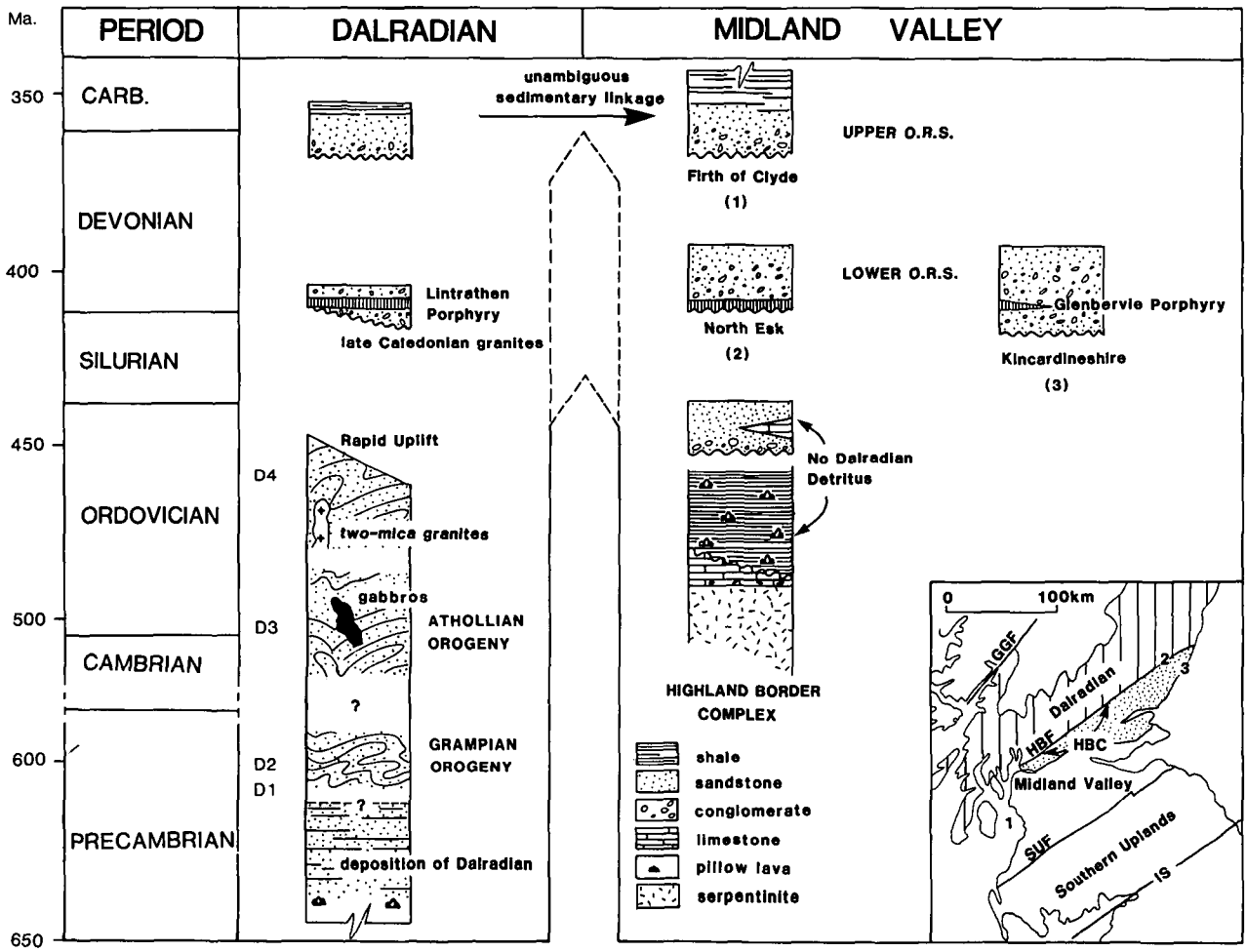


Figure 1. Summary of the geological evolution of the Grampian (Dalradian) and Midland Valley terranes flanking the Highland Boundary Fault, taking chronology for Grampian terrane evolution from Rogers *et al.* (1989) and for Highland Border Complex from Bluck (1985). Inset: geological sketch map of Scotland with major tectonic elements. GGF – Great Glen Fault; HBF – Highland Boundary Fault; HBC – Highland Border Complex; SUF – Southern Upland Fault; IS – Iapetus Suture.

correlation of porphyritic dacites (ignimbrites) of the Lower ORS successions to the north and south of the HBF (Fig. 2). The dacites were interpreted as ignimbrites by Paterson & Harris (1969) following recognition of glass shards and pumice fragments in thin section. The validity of the correlation has important implications for the stratigraphy of the Lower ORS (see Armstrong & Paterson, 1970), and for the scale and timing of possible lateral/orthogonal movements on intervening structures.

Allan (1928) described scattered exposures of a distinctive Lower ORS dacitic porphyry, termed the 'Lintrathen Porphyry', in the Alyth area where it reaches over 150 m in thickness. Paterson & Harris (1969) showed a similar unit to rest unconformably on the Dalradian at West Cult (see Fig. 2). At Glenbervie, in the east, dacitic 'lavas' bearing a close resemblance to those at Lintrathen had been recognized by Campbell (1913), and were correlated by Allan (1928). Here, the 'porphyry' occurs close to the top of the Crawton Group, in the middle of the thick Lower ORS succession of the northeast Midland Valley. Elsewhere, in the North Esk section, a dacite rests with unconformity on the Highland Border Complex (Bluck, 1985). If the correlation of these scattered outcrops is correct, the ignimbrites would link the Dalradian to both the Highland Border Complex and the Midland Valley Lower ORS succession. Thirlwall (1988) reported a recalculated Rb–Sr biotite age for the 'Lintrathen Porphyry' at Loch Lintrathen of 415.5 ± 5.8 Ma, i.e. late Silurian on the timescale of Harland *et al.* (1982). Correlation would therefore constrain the relative positions of the NMV and Grampian terranes at this time.

In this account we report on palaeomagnetic results from porphyries/ignimbrites at two localities: (1) Loch Lintrathen, north of the HBF, where the ignimbrites dip gently northwards, and (2) Glenbervie, south of the HBF, where they crop out on the steep (southward dipping) northwestern limb of the Strathmore syncline. A major unconformity intervenes between the ?Pridoli–Emsian Lower ORS rocks and the late Devonian–Carboniferous Upper ORS. Lower ORS rocks were folded prior to deposition of the

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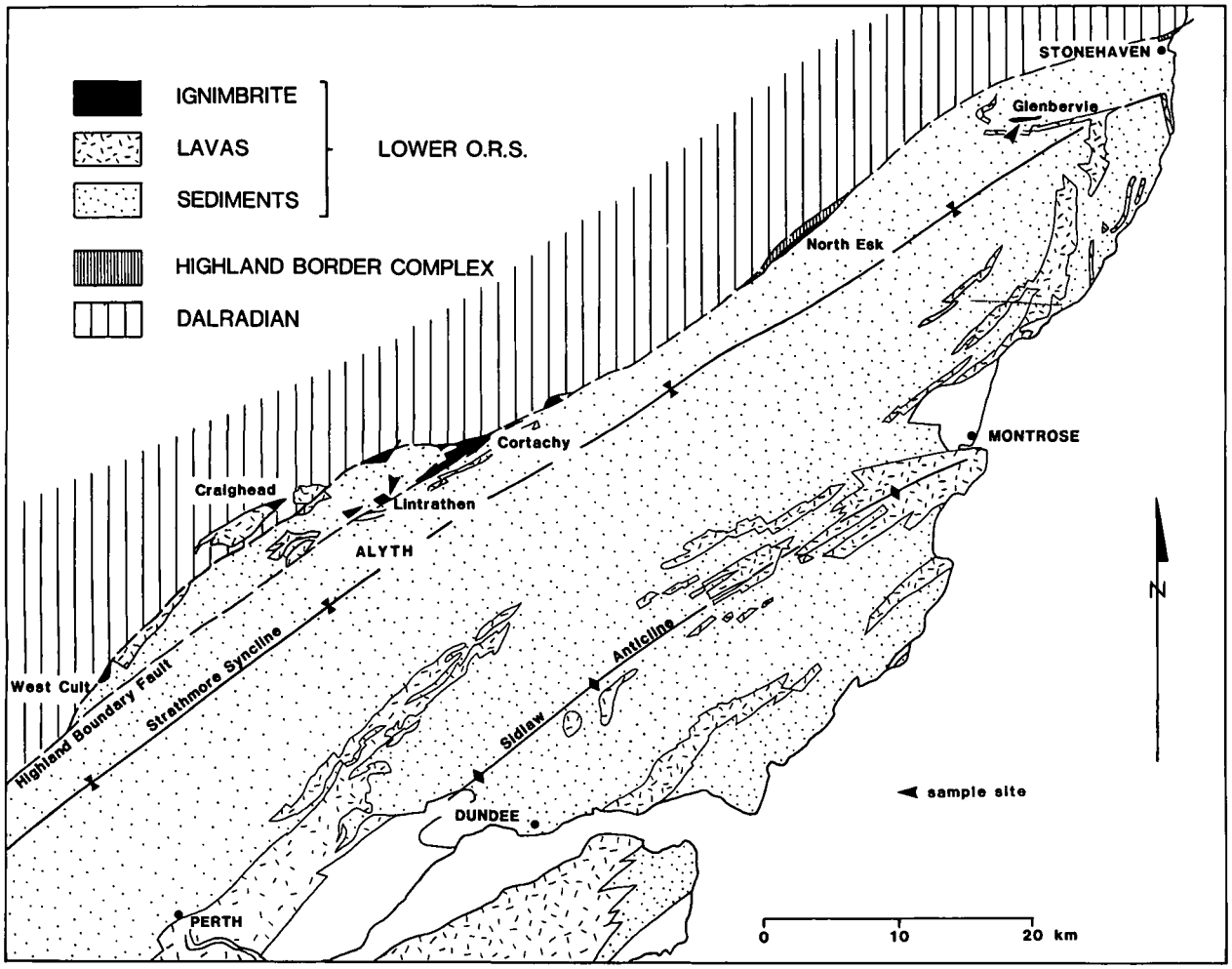


Figure 2. The distribution of Lower Old Red Sandstone ignimbrite exposures along the Highland Border based on Paterson & Harris (1969). Sampling localities at Loch Lintrathen and Glenbervie are indicated to the north and south of the Highland Boundary Fault respectively.

Upper ORS, and this allows the relative ages of the palaeomagnetic components to be deduced using a regional palaeomagnetic fold test.

In the case of movements along a relatively 'straight' fault zone, i.e. a distant euler pole, the palaeomagnetic resolution power is limited and, realistically, displacements less than 500 km cannot be resolved. However, in addition to palaeolatitudinal determinations, remanence polarity, demagnetization behaviour and bulk magnetic properties may provide useful, albeit circumstantial, evidence in assessing the possible correlation of ignimbrites between the two localities.

In this instance, bulk chemical analysis is not a viable method of correlation as the ignimbrites are strongly devitrified and often contain schist and slate country rock fragments. In addition, many ignimbrites are compositionally heterogeneous, with vertical and lateral variations often reflecting a source within a zoned magma chamber. Distinct compositions would not, therefore, necessarily rule out correlation. This problem is further compounded by the lack of a coherent stratigraphy within the ignimbrites/porphyries at either locality. It is thus unclear how many flows are present, although Paterson & Harris (1969) suggest the presence of at least two at West Cult (Fig. 2). Nevertheless, we supplement the palaeomagnetic results with a comparative mineral chemistry study, using microprobe analyses of biotites from the two localities.

Previous palaeomagnetic studies in the area were undertaken by Sallomy & Piper (1973) and Torsvik (1985). The former study included seven sites from three ignimbrite localities at West Cult (3 sites), Loch Lintrathen (3 sites) and Glenbervie (1 site). Three sites were found to be 'normally' magnetized and a fourth site displayed an intermediate polarity. The remaining sites were unclassified.

Torsvik's (1985) study is significant in that modern laboratory procedures were applied to the LORS Midland Valley lavas (although no ignimbrites were sampled). Direct comparison of palaeomagnetic component structure between ignimbrites and lavas is therefore possible in this study.

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2. Palaeomagnetic and rock magnetic sampling and methods

Samples were cored in the field using a portable drilling apparatus. These were subsequently orientated using both magnetic and sun compasses wherever possible. A variably developed primary foliation at each locality was used to constrain the palaeohorizontal. This agreed with bedding measurements in the surrounding sandstones and conglomerates. The foliation is defined by streaks of devitrified glass within the ignimbrite's groundmass and the parallel alignment of biotite flakes and lithic fragments.

The magnetic moments (NRM Natural Remanent Magnetization) of trimmed core samples were measured using a spinner magnetometer (Molyneux, 1971). Magnetic susceptibilities were measured in the field using a Micro-kappa Kappameter Model KT-5. Susceptibility measurements were made wherever possible on flat and unweathered surfaces close to or at the position of sample drilling. This minimized the negative effects of surface roughness. No account was taken of the orientation of the rock surfaces used in susceptibility measurement since Ellwood (1982) reports only modest anisotropies ($< 3.5\%$) for welded tuffs regarded by Rosenbaum (1986) to be generally representative for ignimbrites.

Thermal demagnetization was predominantly employed during the laboratory cleaning of samples due to:

(i) the presence of high coercivity hematite in many samples;

(ii) the introduction of gyroremanent magnetizations (GRM; Stephenson, 1980*a*,*b*) during high field static alternating field (AF) cleaning.

Thermal cleaning was performed using two bench-type thermal demagnetizers controlled by microcomputer. The thermal demagnetizers are each shielded from the ambient Earth's field by four layers of Mu-metal. This reduces the field within the sample chamber to approximately 4 nT.

Alternating field cleaning was performed using a static three-axis system. The demagnetizer is interfaced with a BBC microcomputer which provides a digitally generated ramp multiplied by a pure 186 Hz sine wave. This signal is then sent through an eight-pole Chebychev filter to attenuate harmonics. The demagnetization solenoid is housed within two Mu-metal shields for Earth's field attenuation to less than 15 nT.

Demagnetization results were plotted on orthogonal projections and interpreted interactively using the LINEFIND statistical analysis algorithm (Kent, Briden & Mardia, 1983).

Sites 1–9 were sampled from hillside exposures east of Barhead [NO767811], *c.* 500 m north of Glenbervie. Sites 10–18 were from a disused quarry southwest of Bridgend Farm at Loch Lintrathen [NO280543]. Sites 12 and 15 (both at Loch Lintrathen) were later resampled due to their possessing particular demagnetization complexity and an anomalous direction respectively.

In the field, sites were nominally defined as sample groupings separated either laterally or vertically by several metres of outcrop/non-exposure. With the field recognition of flow/cooling units proving impossible, statistical analysis has been performed at 'sample' and 'locality' rather than 'site' level. Had field recognition of flows been possible, we feel the more correct analysis would be achieved by averaging the results of individual flows.

3. Initial magnetic properties

NRM intensity and volume susceptibility values from both Loch Lintrathen and Glenbervie display approximate log-normal distributions (Fig. 3).

Significant differences in the magnitudes of each property are observed between the two localities. These correspond to a difference in the relative abundance of magnetite phenocrysts observed in thin section. The Glenbervie 'ignimbrite' contains the more abundant magnetite fraction and consequently has a greater susceptibility and NRM intensity than the Loch Lintrathen 'ignimbrite'.

Possible explanations for the different magnetite content are as follows:

(i) The two ignimbrites were derived from compositionally distinct volcanic centres, and may be unrelated.

(ii) The ignimbrites represent proximal and distal parts of a large ash flow sheet or sheets derived from a single volcanic centre with a compositional zonation of magnetite phenocrysts with increasing distance from source. Systematic variation in bulk composition and in the variety and composition of phenocrysts during ash flow eruptions is commonplace (e.g. Smith, 1979). Furthermore, Hahn, Rose & Meyers (1979) show that lateral compositional variation can occur within a single ash flow sheet. They relate this to either more local, late-stage mafic eruptions, or crystal-glass fractionation during emplacement.

(iii) The observed contrasts reflect a vertical stratification of magnetic properties through a single ignimbrite deposit with different levels sampled at the two localities.

4. Progressive demagnetization behaviour

4.a. Glenbervie (south of HBF, sites 1–9)

NRM directions from Glenbervie display south-easterly declinations and moderate negative inclinations (*in situ*). Three magnetization components were revealed upon thermal demagnetization which have been termed components (R), (S) and (P). They are distinguished by their respective directions of mag-

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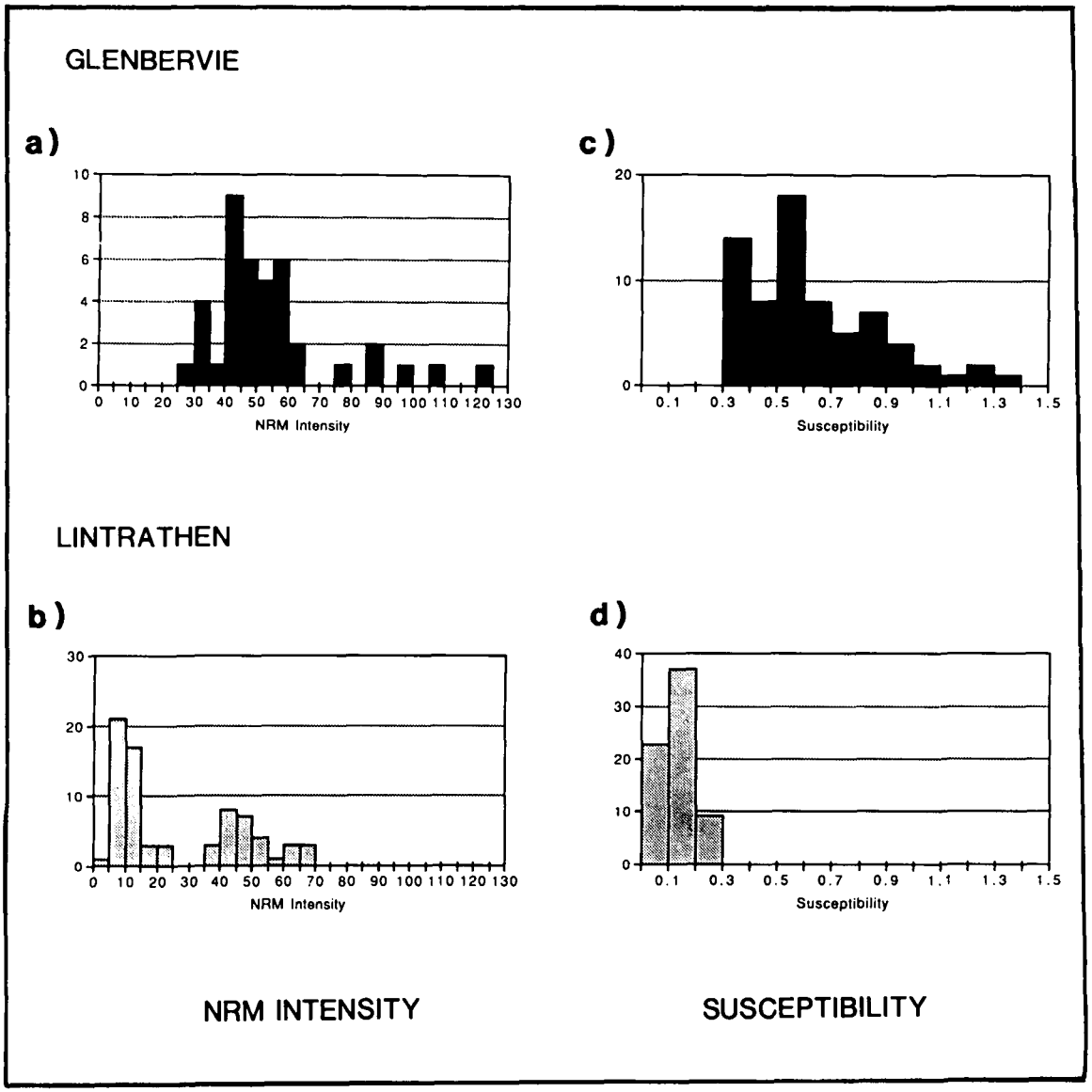


Figure. 3. Initial magnetic properties for the Glenbervie (a, c) and Loch Lintrathen (b, d) ignimbrite localities. (a, b) NRM intensity: milliAmps/metre. (c, d) Apparent Volume Susceptibility: S.I. units $\times 10^{-3}$. Parameters are plotted on a linear scale.

netization and unblocking spectra as follows:

4.a.1. Component (R)

Component (R) is readily removed during laboratory cleaning, usually unblocking below 200 °C and at coercivities less than 10 mT (Fig. 4). Overlap of blocking spectra with component (S) is occasionally noted, extending the stability range to a maximum of 250–300 °C. *In situ*, (R) has a steep positive inclination and a generally northward declination subparallel to the Earth's present field at the locality. We interpret this component as a recent viscous remanent magnetization (VRM).

4.a.2. Component (S)

Component (S) unblocks at temperatures of between 200 and 450 °C. *In situ*, it has a southerly declination and moderate negative inclination. Occasional overlapping stability with (R) has been noted above (see 4.a.1). Similarly, overlap with component (P) is considerable in some samples whilst negligible in others (compare Fig. 4c and 4a). Component (S) can be identified in most thermally demagnetized samples but its direction is not always determinable given the spectra overlap and ensuing 'curved' demagnetization trajectories.

4.a.3. Component (P)

Component (P) unblocks between 450 and 650 °C, decaying directly to the origin (Fig. 4). *In situ*, it has southeasterly declination and moderate–steep negative inclination. It is present in all samples and usually forms the dominant magnetization component.

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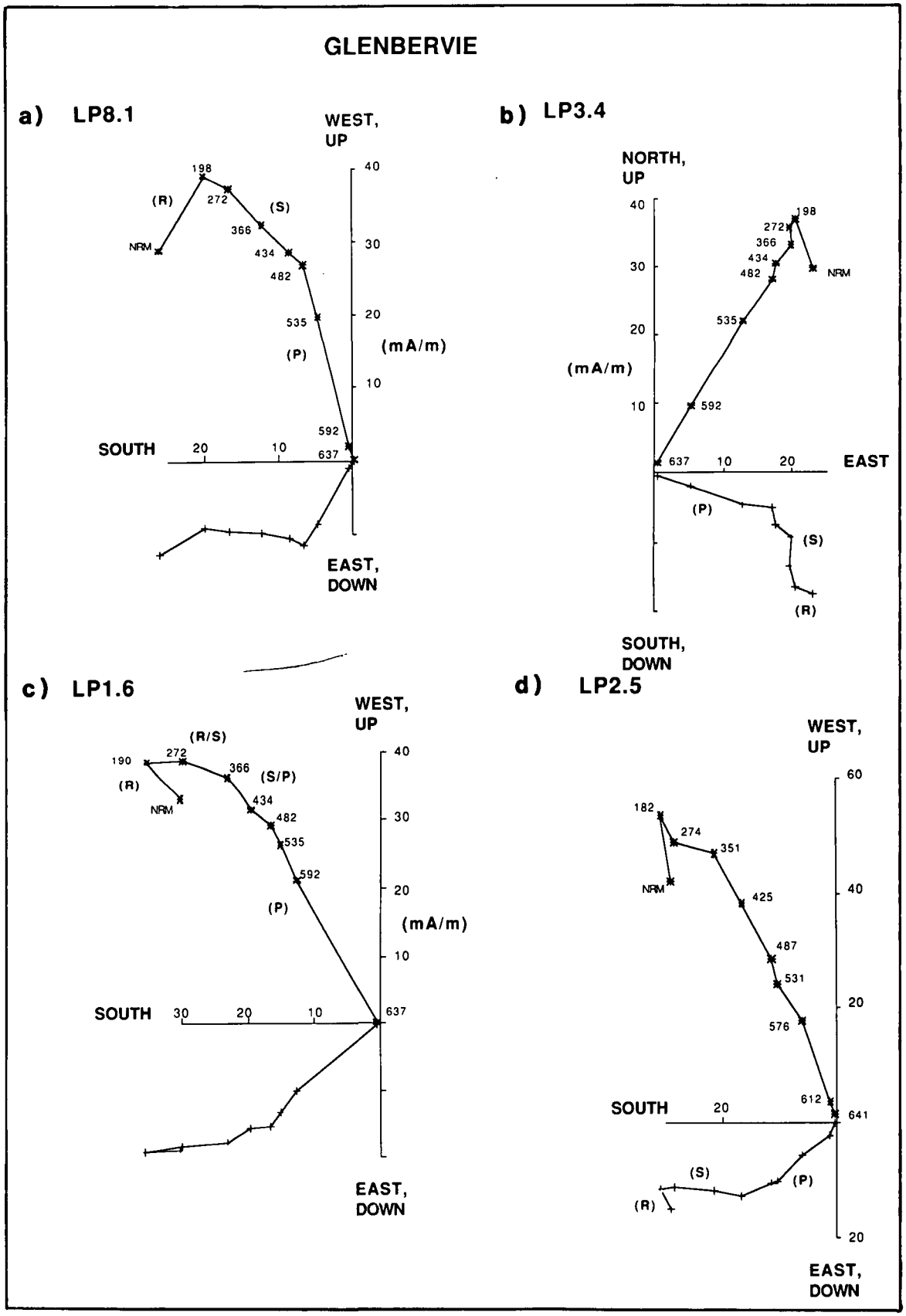


Figure 4. Representative orthogonal projections for the Glenbervie ignimbrite. Components of NRM are labelled. In the orthogonal vector diagrams crosses (stars) represent points in the horizontal (vertical) plane. (a, b, d) Three-component structure (R, S and P) with little overlap of blocking spectra. (c) The presence of three components is determinable but considerable overlap of unblocking spectra is evident.

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4.b. Loch Lintrathen (north of HBF, sites 10–18)

In situ NRM directions from the Loch Lintrathen ignimbrite have northeasterly declinations and shallow positive–negative inclinations. Three main magnetization components were again identified during thermal cleaning. The presence of a fourth component was suspected at site 12 and this was later confirmed by resampling and further analysis (site 12A; Fig. 5c, d). A similar component nomenclature to that developed for the Glenbervie 'ignimbrite' is adopted as the demagnetization patterns observed for the two localities are very similar. The additional component identified at site 12/12A was termed component (D).

4.b.1. Component (R)

Component (R) was removed at temperatures below 200 °C. *In situ*, the component is northerly directed with steep positive inclinations (Fig. 5a, b, e) and is regarded (analogous to Glenbervie) to represent a recent VRM. Overlap with (S) (when present) again occasionally extends the unblocking range to 250–300 °C.

4.b.2. Component (S)

Component (S) was revealed as a linear segment only at sites 12/12A and in two samples from site 17. When observed, the component unblocks at temperatures between 200 °C and 450 °C and is southerly directed (*in situ*) with moderate negative inclinations. Its presence within other samples and at other sites could be deduced from a 'looping' of the orthogonal projection over the appropriate blocking temperature range (Fig. 5e, also site 17). No accurate determination of the component orientation proved possible in these circumstances however.

4.b.3. Component (P)

Component (P) is northeasterly directed having shallow–moderate negative inclination (*in situ*). It unblocks at temperatures in excess of 450 °C and decays directly to the origin by 670 °C (Fig. 5e, f). For site 12/12A however, the component has a restricted unblocking range between 450 and 630 °C (Fig. 5a, b) and does not decay to the origin (Fig. 5c, d).

AF demagnetization of samples from sites 10 and 14 was comparable to thermal treatment for these sites. Component (P) determinations (three results) were therefore included within the overall statistical mean. An AF demagnetization from site 17 revealing (R) and (P) was omitted, however, as linear component (S) segments were revealed on thermal treatment of two samples from the site.

4.b.4. Component (D)

Component (D) was present only at sites 12 and 12A (although it may be present in a sample from site 10). It unblocks exclusively in the 640–670 °C range and is therefore considered to reside within hematite. Although revealed only in the final stages of demagnetization, detailed analyses of samples from site 12A prove it to be reproducible (Fig. 5a–d). *In situ*, the component has a NE–NNE declination and shallow positive inclination (Fig. 6a), becoming negatively inclined upon structural correction (Fig. 6b; Table 1). The lack of a comparable component for the Glenbervie ignimbrite precludes the inclusion of component (D) in a fold test.

The age and origin of (D) is briefly discussed here although, given its limited occurrence, conclusions must be regarded as tentative.

The similarity in pole position of component (D) with results reported from Middle–Upper Devonian rocks throughout Scotland is noted (e.g. Storetvedt & Torsvik, 1983, 1985), with agreement between poles being marginally better after tectonic correction. The improvement is not statistically significant, however, as both the *in situ* and corrected poles overlap with the best fit APWP (Torsvik *et al.* 1990 and Fig. 9).

Torsvik *et al.* (1989) suggest an early Carboniferous age for similar magnetizations recorded in ORS rocks from the western Midland Valley based upon negative fold and conglomerate tests. They consider overprinting to be 'thermo-chemical', relating to a high geothermal gradient during extrusion of the Lower Carboniferous lavas.

The component (D) orientation at Loch Lintrathen contradicts that expected should the Storetvedt (1987) model to account movements along the Great Glen Fault (GGF) be considered (i.e. the D pole data agrees with equivalent results to the *northwest* of the GGF). We therefore opine that the tectonic solution given by Storetvedt (1987) is incorrect.

4.b.5. Anomalous directions from site 15

NRM and demagnetization results from site 15 did not conform to the component structure outlined above (sections 4.b.1–4.b.4). A minor (steep inclination) component removed below 200 °C gives way to a magnetization of southeasterly declination and moderate negative inclination. This component is thermally discrete and unblocks at temperatures above the Curie point of magnetite.

Site 14, positioned at a stratigraphic level 2 m below site 15, displays a 'normally directed' (northeast declination, negative inclination) thermally discrete component (P) (Fig. 5d).

Sites 14 and 15 were reoccupied and a traverse between them sampled for study. An abrupt change in the NRM orientation (and of the thermally discrete

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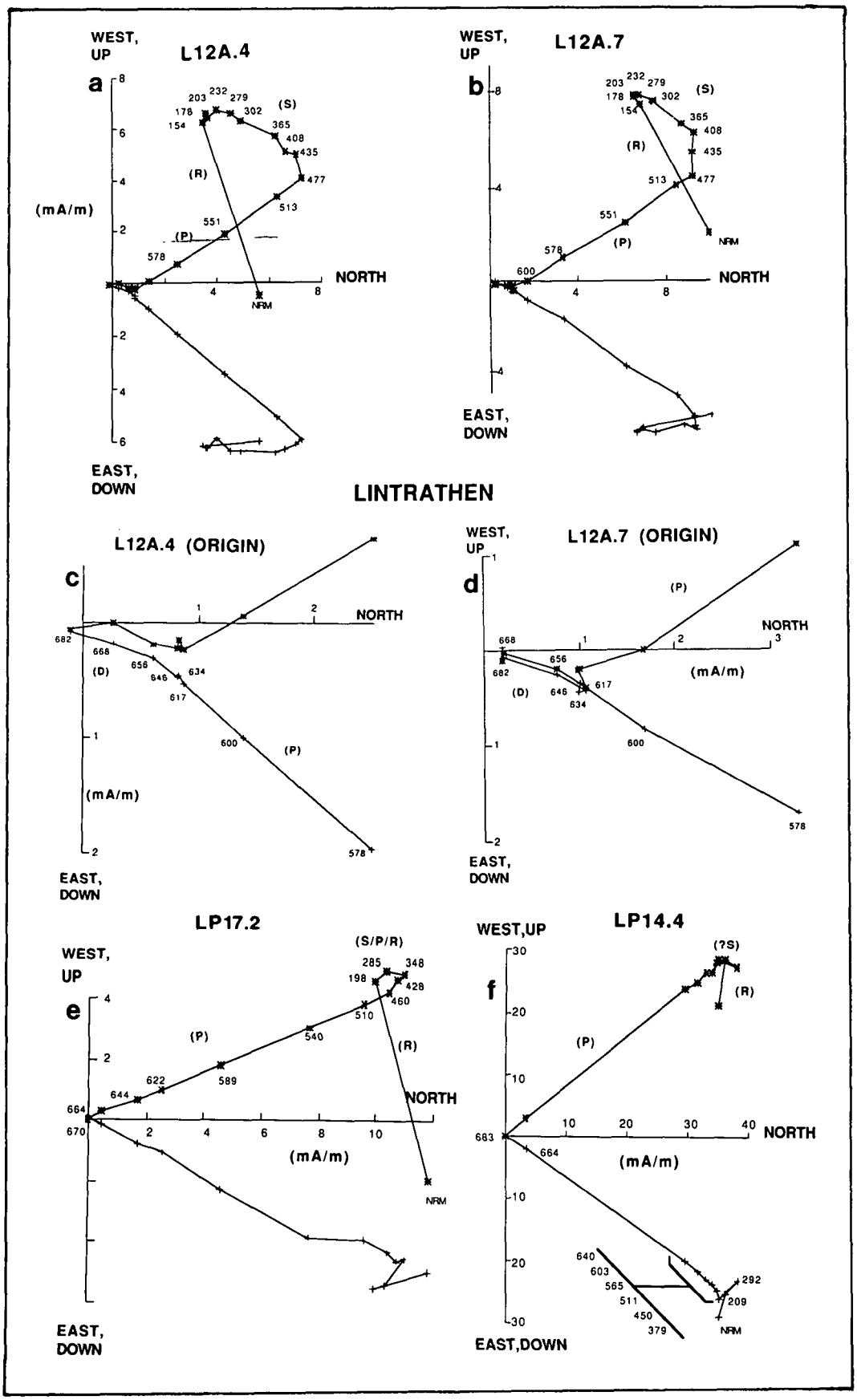


Figure 5. Representative orthogonal projections for the Loch Lintrathen ignimbrite. Interpreted components of NRM have been labelled. (a, b) Four-component demagnetization (R, S, P, D) for site 12. Projections are expanded near the origin (c, d) to show component (D). (e) Predominant component (P) at site 17. Presence of a latent component (S) is indicated by a 'looping' of the orthogonal projection in the range 200–400 °C. (f) A thermally discrete component (P) dominates the NRM at site 14.

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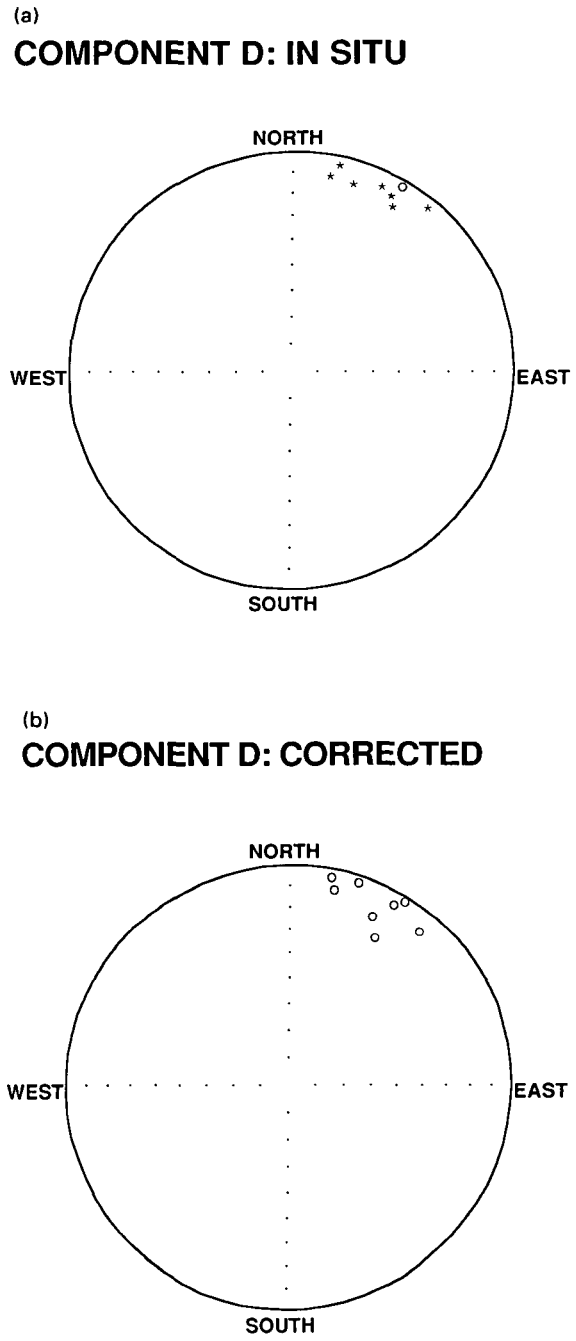


Figure 6. *In situ* (a) and structurally corrected (b) component (D) distributions for the Loch Lintrathen ignimbrite. Stars (open circles) represent points in the lower (upper) hemisphere.

component) was found to occur between the sites. This coincided with a similarly abrupt swing in the attitude of the weak mineral alignment foliation identified in the recovered cores.

As we were unable accurately to determine the new alignment of this foliation, site 15 is excluded from further discussion and from the overall statistical analyses.

5. Fold tests

The contrasting attitude of the ignimbrite foliation at Loch Lintrathen and Glenbervie and the identification of equivalent remanence components allows the performance of regional fold tests for components (S) and (P). Component (R) forms a recent VRM and would fail a fold test. The fold tests constrain the magnetization age relative to the post-Emsian, pre-Upper ORS deformation. This is assumed to have produced the structural tilt at both localities. Structural correction was performed about a horizontal strike axis at each locality.

5.a. Component (S)

In situ and structurally corrected distributions of component (S) for both ignimbrites (Fig. 7; Table 2) suggest the component to post-date tectonism at the 99% confidence level (McElhinny, 1964). *In situ* (S) poles are statistically equivalent at 95% confidence and are comparable with several remagnetization poles reported from the Scottish ORS (termed 'Group A' poles; Tarling, 1985). Several magnetic ages have been proposed for these results spanning Permo-Carboniferous to Tertiary times (Cisowski, 1984). Given the overall scatter of the poles, we suggest a conservative age estimate of late Palaeozoic–early Mesozoic. The poles are plotted on a combined Torsvik *et al.* (1990) and Jowett, Pearce & Rydzewski (1987) APWP in Figure 9.

The (S) poles agree with equivalent data *northwest* of the Great Glen (in the context of the Storetvedt (1987) GGF model) despite being derived from rocks exposed to the *southeast* of the fault. This once again contradicts Storetvedt's proposed tectonic solution (see also section 4.b.4).

Table 1. Component (D) Fisher statistics *in situ* and structurally corrected

Sample location	*N*	Dec	Inc	*k*	α95	Pole Long.	Pole Lat.	dP	dM
In situ									
Lintrathen	8	24.1	9.8	45.2	8.3	327.0 E	35.0 S	4.2	8.4
Corrected									
Lintrathen	8	24.0	−10.9	45.2	8.3	330.2 E	24.9 S	4.2	8.4

N, number of samples; *k* – Fisher precision parameter (Fisher, 1953); Dec – palaeomagnetic declination; Inc – palaeomagnetic inclination; α95 – cone of 95% confidence about the mean; dP, dM, semi-axes of the oval of 95% confidence about the pole.

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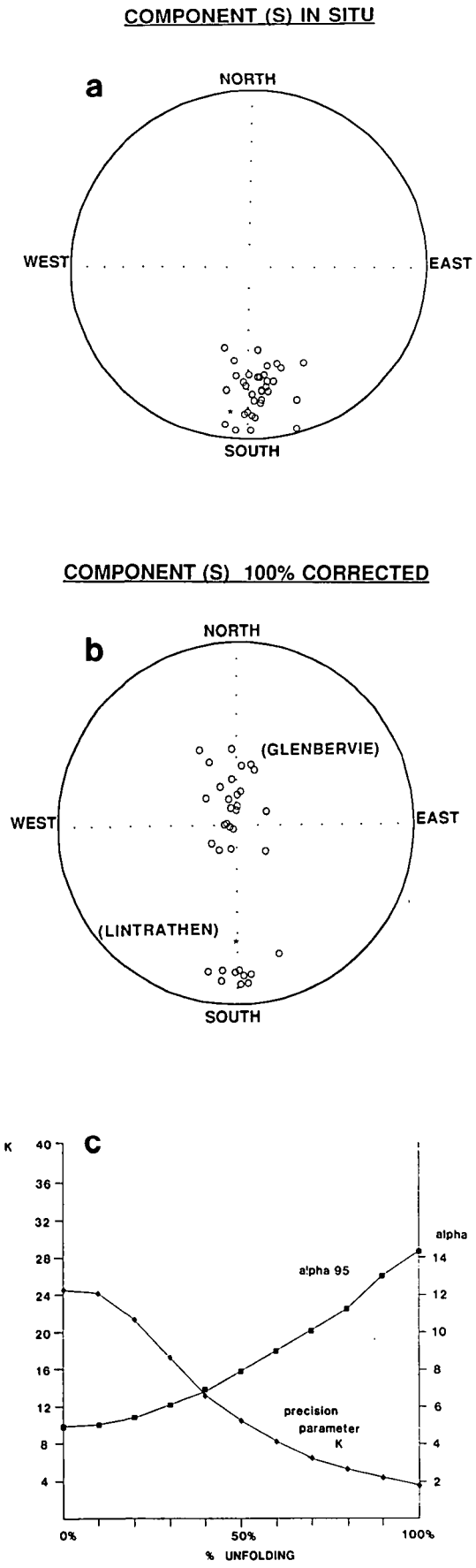


Figure 7. *In situ* (a) and structurally corrected (b) (S) components from the Loch Lintrathen and Glenbervie ignimbrites. (c) Variation of Fisher (1953) statistical parameters with incremental structural tilt correction.

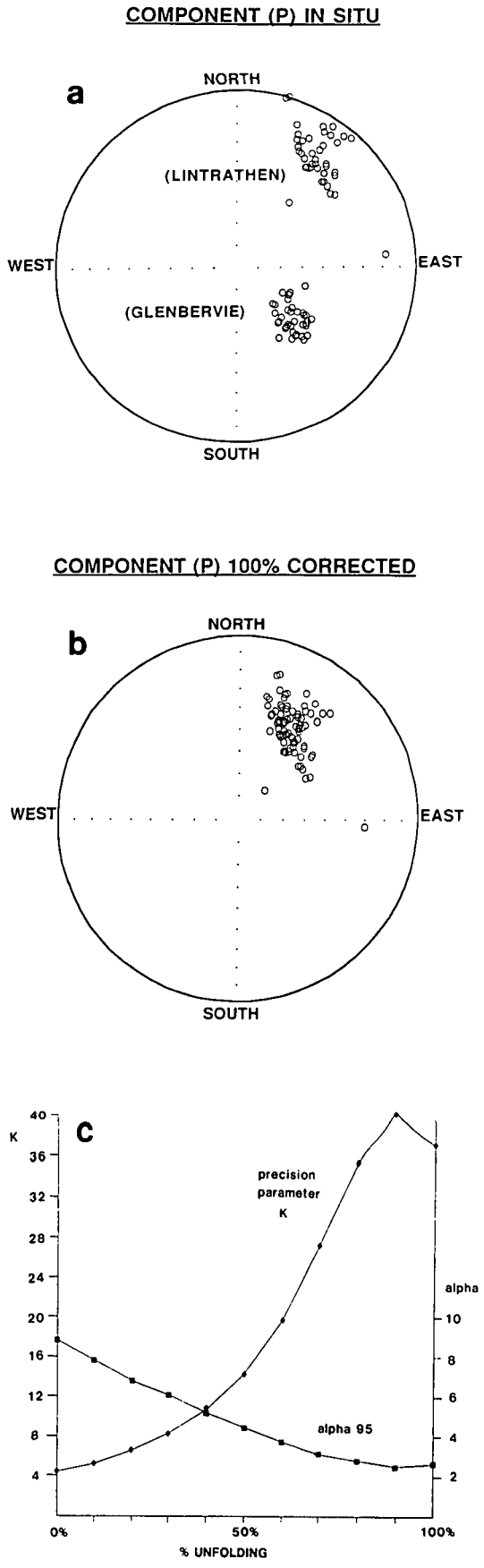


Figure 8. *In situ* (a) and corrected (b) component (P) directions for the Loch Lintrathen and Glenbervie ignimbrites. Anomalous (probable) component (P) directions from site 15 are omitted. (c) Fisher (1953) statistics with incremental structural correction.

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Table 2. Component (S) Fisher statistics *in situ* and structurally corrected for both Lintrathen and Glenbervie, together with combined mean results

Sample location	*N*	Dec	Inc	*k*	α95	Pole Long.	Pole Lat.	dP	dM
In situ									
Glenbervie	24	177.8	−25.7	25.9	5.9	0.7 E	46.7 S	3.4	6.4
Lintrathen	11	175.4	−29.4	20.7	10.3	3.5 E	49.0 S	6.3	11.4
Combined	35	177.1	−26.9	24.5	5.0	2.1 E	47.9 S	3.0	5.4
Corrected									
Glenbervie	24	341.6	−79.9	25.9	5.9				
Lintrathen	11	180.4	−12.6	20.7	10.3				
Combined	35	192.0	−25.4	3.7	14.6				

Glenbervie strike/dip = 85.2/74.1 (S). Lintrathen strike/dip = 305.6/21.1 (N)
Table parameters are as for Table 1. Strike and dip values shown were utilized in the structural correction of results.

Table 3. Component (P) Fisher statistics *in situ* and structurally corrected for Lintrathen and Glenbervie together with combined mean results

Sample location	*N*	Dec	Inc	*k*	α95	Pole Long.	Pole Lat.	dP	dM
In situ									
Glenbervie	36	134.6	−54.7	96.2	2.5				
Lintrathen	41	36.1	−24.4	28.5	4.3				
Combined	77	67.6	−49.6	4.4	8.7				
Corrected									
Glenbervie	36	24.7	−40.2	96.2	2.5	334.7 E	7.6 S	1.8	3.0
Lintrathen	41	36.1	−45.6	28.5	4.3	325.1 E	1.0 S	3.5	5.5
Combined	77	30.4	−43.2	37.1	2.7	329.9 E	4.3 S	2.1	3.3
	Palaeolatitude			dP (uncertainty on palaeolatitude)					
Glenbervie	22.9° S			1.82°					
Lintrathen	27.0° S			3.48°					
Combined	25.2° S			2.10°					

Calculated palaeolatitudes and their corresponding 95% uncertainties are also shown. Table parameters are as for Table 1. Strike and dip values are as Table 2.

5.b. Component (P)

The structurally corrected (P) distribution (Fig. 8; Table 3) groups significantly better at 99% confidence than its *in situ* counterpart (McElhinny, 1964) and suggests a pre-tectonic origin. The component is interpreted to represent a primary magnetization acquired at or close to the time of ignimbrite emplacement.

A local peak in precision at 90% unfolding (Fig. 8c) is not taken to indicate a syn-folding origin for two reasons:

(*a*) The syn-folding peak does not approach statistical significance (95% confidence) when compared to the fully corrected data.

(*b*) Greatest precision close to complete unfolding may indicate the palaeo-horizontal to have been incorrectly located by a few degrees at either locality. Syn-depositional dips of several degrees are commonplace within volcanic successions.

Determined palaeolatitudes of ignimbrite emplacement calculated from component (P) are not distinguishable across the Highland Boundary at 95% confidence (Table 3, Loch Lintrathen 27.0° S, Glenbervie (22.9° S). There is therefore no palaeomagnetically detectable post-Lower ORS (late Silurian) translation along the HBF. The combined palaeolatitude for extrusion is calculated at 25.2° S ± 2.1°. However, a marginally significant declination difference causes the two virtual geomagnetic poles (VGPs) to be distinguishable across the HBF (Table 3; Fig. 9). Nevertheless, the VGPs confirm the late Silurian–early Devonian 'corner' in the Scottish Palaeozoic apparent polar wander path (Briden, Turnell & Watts, 1986; Torsvik *et al.* 1990), which is suggested to broadly constrain the docking of the Grampian Highlands, Midland Valley and Southern Uplands terranes at this time (Trench *et al.* 1989).

A contrast in precision between the respective component (P) distributions for each locality is evident in Table 3 (Loch Lintrathen, $k = 28.5$; Glenbervie, $k = 96.7$). This may result from sampling the greater stratigraphic thickness at Loch Lintrathen, perhaps incorporating multiple cooling units. Precision decrease could then reflect a longer time-averaging of the ambient field. Decrease in precision with increasing unit thickness has been reported by Rosenbaum (1986) within Miocene tuffs of southern Nevada.

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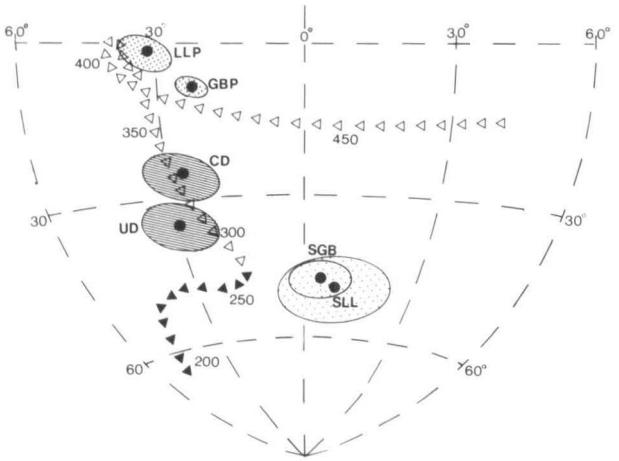


Figure 9. Pole positions and shaded 95% confidence limits for components (P), (S) and (D) plotted in equal area projection. Open triangles trace the locus of the northern British APWP determined north of the Iapetus suture but south of the Great Glen Fault (from Torsvik *et al.* 1990). Infilled triangles denote an extension of this path based on a compilation of late Palaeozoic and early Mesozoic European results (from the 20 Ma path of Jowett, Pearce & Rydzewski, 1987). Age calibration is approximate. Poles are as follows: LLP, GBP – corrected Lintrathen and Glenbervie component (P) poles respectively; UD, CD – *in situ* and corrected component (D) poles respectively; SLL, SGB – *in situ* Lintrathen and Glenbervie component (S) poles respectively.

6. Gyroremanent and anhysteretic acquisition of magnetization

All samples subjected to standard three-axis AF cleaning were found to acquire a 'spurious' magnetization at treatments in excess of 35–40 mT. This was manifest as an 'oscillatory' or 'spiralling' movement in orthogonal projection which varied systematically with the final sample demagnetization axis.

An empirical method for the removal of this 'axis dependent' contamination was proposed by Dankers & Zijderveld (1981). A theoretical basis for its generation was developed by Stephenson (1980*a*,*b*, 1981) in terms of the gyroscopic properties of the sample moments; the acquired remanence is referred to as a Gyroremanent Magnetization (GRM).

The Dankers & Zijderveld (1981) method for GRM removal was adopted and proved successful although was not adopted extensively due to the success of thermal treatment in separating components.

GRM acquisition in ignimbrites at both Loch Lintrathen and Glenbervie constitutes circumstantial evidence in favour of their proposed correlation. Langereis (1979) notes that 'the introduction of disturbing remanences of this type (GRM) in higher alternating fields *very rarely* occurs in palaeomagnetic research'. Whilst lack of recognition no doubt accounts for some of this 'rarity', we note that GRM is easily recognized at both localities.

GRM has been linked to magnetically anisotropic single domain magnetite (Stephenson, 1981), which is thus inferred to be present in the ignimbrites. Magnetic anisotropy (presumably of only a few per cent; Rosenbaum, 1986) is likely to relate to the plane of ignimbrite foliation, although this has not been directly investigated in the present study.

7. Biotite chemistry

The chemistry of biotites in five samples (hand samples and offcuts from sample cores) from each locality has been determined by EDS electron microprobe. Compositions are summarized in Figure 10. Biotite in pyroclastic flows is particularly susceptible to pre- and syn-eruptive fluid exchange. Magmatic compositions may therefore not be retained. However, Hildreth (1979) shows that magmatic octahedral compositions were generally preserved in biotites from the Bishops Tuff of eastern California.

Mg, Fe and Ti abundances for biotites from Loch Linthrathen and Glenbervie have a narrow and similar range (Fig. 10a, b). This is consistent with, but does not establish, a source in the same volcanic centre, accompanied by a similar history of alteration during eruption, emplacement and cooling.

8. Discussion

Statistically equivalent palaeolatitudes determined for the Loch Lintrathen and Glenbervie 'ignimbrites' require that any post-ignimbrite net translation along the HBF or its precursors did not exceed the palaeomagnetic errors (Table 3). This constraint applies largely to strike-slip movement, given the similarity of the palaeo-azimuth with the present trend of the HBF.

This confirms the previously noted equivalence of palaeomagnetic data from the Lorne Plateau and Midland Valley ORS lavas across the terrane boundary (Harte *et al.* 1984). As noted above, the ignimbrite (P) poles are statistically separated due to a marginal declination difference between the localities. This may either reflect a problem with the method of structural correction (see MacDonald, 1980) or a true tectonic rotation analogous to that recorded within Wenlock redbeds in western Ireland (Smethurst & Briden, 1988).

Given that the palaeomagnetic data alone cannot exclude modest potential movements between blocks, i.e. up to approximately 500 km, we explore the evidence for direct correlation of the 'ignimbrites' north and south of the HBF. This might allow a tighter constraint on the relative late Silurian position of the NMV and Grampian terranes.

In this respect, we note that several demagnetization properties of the Loch Lintrathen and Glenbervie ignimbrites are found to be unusual but directly

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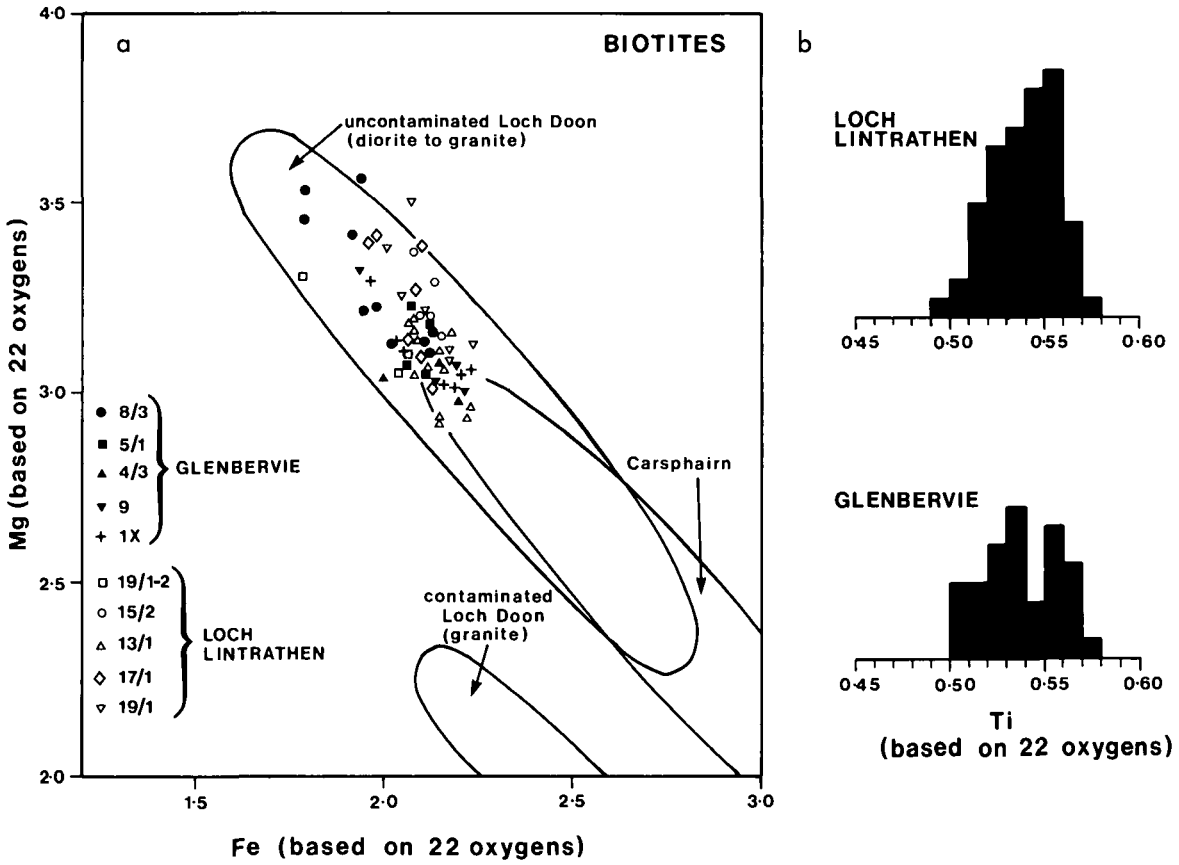


Figure 10. Biotite data for samples from Loch Lintrathen and Glenbervie. (a) Mg versus Fe plot (based on 22 oxygens) with data superimposed on fields for biotites in some late Caledonian intrusions (data from Tindle, McGarvie & Webb, 1988). (b) Histograms of Ti content of micas at Loch Lintrathen and Glenbervie.

comparable. These are notably the occurrence of component (S) in both ignimbrites and their similar acquisition of GRM during AF treatment.

These features support the case for correlation between the two localities:

(i) Component (S) 'type' magnetizations have not been found during the detailed thermal demagnetization of surrounding volcanics of the Crawton and Arbuthnott groups (Torsvik, 1985). This component is therefore unique (amongst the Lower ORS volcanics) to the two ignimbrite localities examined.

(ii) GRM acquisition has been reported only occasionally during the routine AF demagnetization of rocks.

The common 'normal' polarity (component (P), Fig. 8) and consistency of palaeolatitude determinations also favour correlation. The common polarity becomes significant given the prevalent occurrence of reversals within the Lower ORS volcanics (Sallomy & Piper 1973; Torsvik, 1985). The comparable biotite chemistry, similar field appearance and petrography (Paterson & Harris, 1969) lend further weight to the correlation.

Significant differences exist, however, in the initial magnetic properties (i.e. NRM intensity, susceptibility). Also, component (D) is only recognized locally at Loch Lintrathen and appears to be absent at Glenbervie. This may result either from insufficient sampling at Glenbervie or reflect the masking of the component potentially by blocking spectra overlap.

We conclude that the weight of the evidence favours the proposed correlation of the units, accepting that there was probably important lateral and/or vertical segregation of magnetite within the deposit. Consequently, net displacements along the HBF which postdate ignimbrite emplacement cannot have exceeded the former scale and geometry of the deposit.

It is possible to explore further the constraints on the timing and likely scales of displacement imposed by the correlation, by considering the possible scale of the original ignimbrite deposit. The distance travelled from source of some typical ignimbrite flows is shown in a histogram in Figure 11a (data from Cas & Wright, 1987). The bulk of these extend no more than 100 km from source, with one example extending to 225 km (the Morrinsville ignimbrite). If the Loch Lintrathen and Glenbervie 'ignimbrites' represent part of the same deposit, the two localities must have lain within the maximum ignimbrite travel distance. Taking a mean travel distance of 56 km, and a source for the ignimbrites close to Loch Lintrathen (the location of the source is unknown, but the maximum thicknesses are preserved here), the relative position of the Grampian and NMV terranes do not have to be

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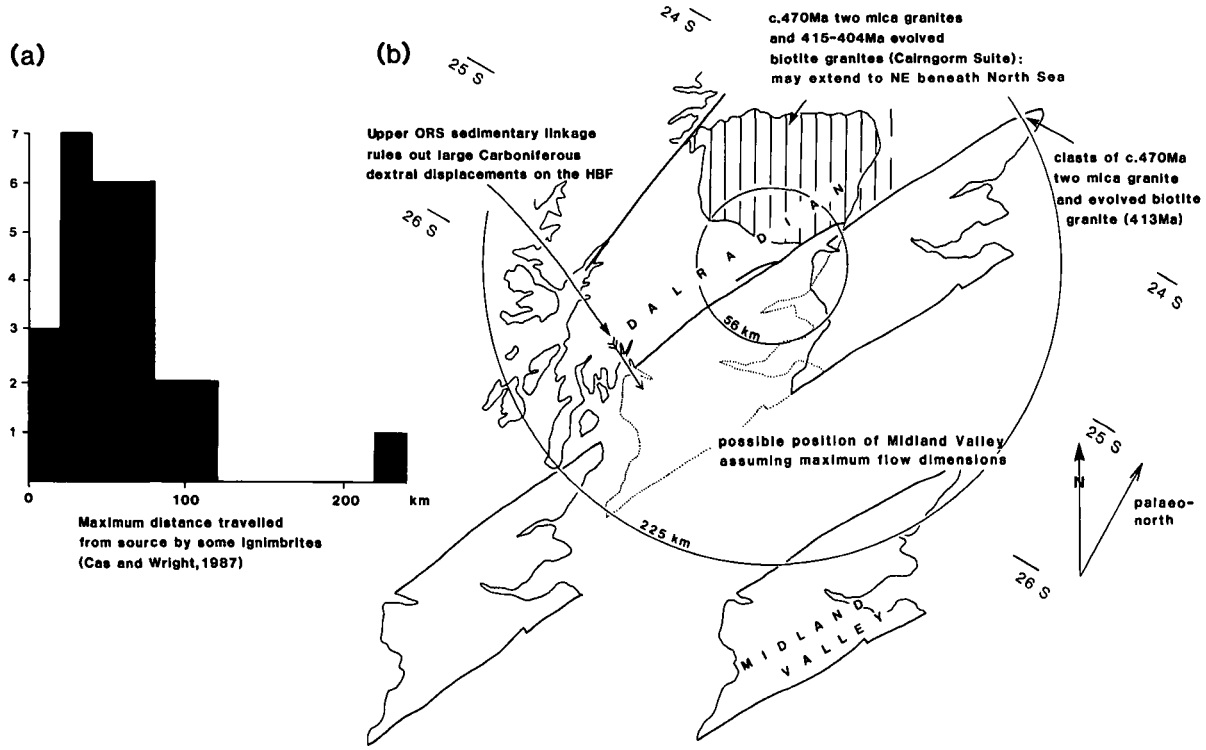


Figure 11. Possible positions of the Midland Valley with respect to the Grampian Highlands given accepted ignimbrite correlation between terranes. (a) A histogram of distances travelled from source by some ignimbrites is shown constructed from data compiled by Cas & Wright (1987). Mean distance (56 km) and maximum distance (225 km) have been used to control possible reconstructions in (b). Note that erosion makes many of the observed travel distances underestimates. Palaeonorth and palaeolatitudes are based on combined Loch Lintrathen and Glenbervie corrected component (P) directions.

greatly different than they are today (Fig. 11b). Accepting the possibility of a longer travel distance, e.g. approaching the 225 km maximum observed distance, Glenbervie (and the NMV) can be positioned at various points around the circumference of a maximum travel distance circle centred on an assumed source close to Loch Lintrathen (Fig. 11b). The possible positions can then be combined with other geological observations to achieve a consistent model.

The provenance of northerly derived boulder conglomerates occurring in the Lower ORS of the northeast Midland Valley provides an important constraint on the relative position of the Grampian and NMV terranes. Haughton, Rogers and Halliday (1990) show that these conglomerates contain large granitoid clasts of two main types: a group of Llanvirn–Caradoc peraluminous two-mica granites and a diverse suite of predominantly red, highly evolved biotite granites (*sensu stricto*) and related porphyries. The granite clasts are shown chemically and isotopically to resemble intrusions with similar ages in a restricted area of the Grampian terrane (Fig. 11b). Unfortunately, it is not clear to what extent this distinctive granite province extends to the northeast beneath the North Sea, and thus it is possible to displace the NMV northeastwards by an unconstrained amount whilst retaining the distinctive clast provenance. Indeed, there is some evidence (discussed in Haughton, Rogers & Halliday, 1990) in favour of a source other than the now exposed part of the northeast Grampian terrane. By combining the granite provenance data with the constraints imposed by this study, we are able to place a limit on the possible eastward (with respect to the Grampian terrane) position of the NMV during late Silurian time (Fig. 11b). With an ignimbrite source close to Loch Lintrathen, and a maximum ignimbrite travel distance of 225 km, the NMV can have been offset left-laterally up to 160 km. This would require a substantial pre-Upper ORS dextral displacement on the HBF to produce the present geometry. Note, however, that this is a maximum estimate and it is possible to satisfy the ignimbrite and granite clast data with smaller offsets.

Data from detrital garnet compositions from the top of the Lower ORS succession (Haughton & Farrow, 1989) suggest that the Grampian and NMV terranes were within tens of kilometres of their present position in Siegenian–Emsian times. Thus the evidence accrued from several independent approaches to constraining the positions of the Grampian and NMV terranes concur in requiring little or no relative motion since late Silurian times.

Given the evidence for Ordovician separation of the terranes summarized in Bluck (1984, 1985), convergence (probably by strike-slip motions; Harte *et al.*

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1984) between the terranes must have taken place between Caradoc–Ashgill (the latest depositional age for Highland Border Complex sediments which contain no Grampian terrane detritus) and late Silurian time. Although the two terranes were probably in close proximity by the late Silurian period, the boundary between them still remained active until Carboniferous times, and may have accommodated small (tens of kilometres) lateral or orthogonal displacements.

One problem in trying to constrain the relative position of the NMV and Grampian terrane is the possibility of two or more phases of displacement with opposing sense. For example, sinistral post-Lower ORS displacements could have continued to drive the NMV eastwards of the Grampian terrane, with a subsequent phase of dextral displacement 'replacing' some or all of the offset, thus obscuring important post-Lower ORS terrane motions. This sort of scenario is made all the more plausible by evidence for regional Acadian (climaxing in the Emsian) sinistral displacements over much of Britain (Soper, Webb & Woodcock, 1987), and the possibility of Carboniferous dextral displacements in the Midland Valley (e.g. Read, 1988). However, the fact that late Devonian–early Carboniferous Upper ORS conglomerates in the western Midland Valley have a source in the 'local' Grampian terrane (Bluck, 1984) suggests that the HBF was not the locus of large post-Upper ORS lateral displacements and consequently we reject complex displacement histories of this type.

9. Conclusions

A palaeomagnetic study of late Silurian dacitic ignimbrites at Glenbervie (south of the HBF) and Loch Lintrathen (north of the HBF) reveals a complex history of magnetization and remagnetization.

At Glenbervie, three magnetization components are recognized, termed (R), (S) and (P), and are attributed Recent, late-Palaeozoic to early-Mesozoic, and late Silurian magnetic ages respectively. At Loch Lintrathen, equivalent components (R), (S) and (P) are revealed, together with a local fourth component (D). This latter component is attributed an early Carboniferous age.

Component (P) (component (S)) passes (fails) a fold test with respect to post-Lower ORS, pre-Upper ORS deformation. Both 'ignimbrites' show comparable gyroremanence upon static AF demagnetization, and the primary magnetization (P) is 'normal' in both cases. Microprobe analyses of biotite phenocrysts from either locality are directly comparable. Palaeolatitudes determined for each locality are statistically indistinguishable across the intervening HBF (combined 25.2° S), limiting substantial post-Lower ORS movements. The data presented for each locality favour their proposed correlation. This consequently allows a more precise constraint to be placed upon the timing of displacements along the HBF or its precursors.

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